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Efficiency of a technology-assisted nutritional screening system: A retrospective analysis of 11,722 admissions in a tertiary hospital

Frank Carrera-Gil ^{a, b, *, 1}, María Isabel Prieto Rusca ^{c, **}^a Servicio de Nutrición Clínica, Fundación Valle del Lili, Cali 760026, Colombia^b Departamento de Alimentación y Nutrición, Facultad de Ciencias de la Salud, Pontificia Universidad Javeriana Seccional Cali, Cali 760021, Colombia^c Departamento de Medicina Interna, Fundación Valle del Lili, Cali 760026, Colombia

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SUMMARY

Background and aim: Nutritional screening is essential for addressing malnutrition and its consequences. However, routine implementation in large hospitals faces several challenges. To overcome these obstacles, the Clinical Nutrition Service of a tertiary hospital developed a technology-assisted nutritional screening system. This study evaluates the system's efficiency in detecting and assessing patients at nutritional risk upon hospital admission. It also examines the association between nutritional risk, clinical outcomes, and sociodemographic characteristics.

Methods: This retrospective, analytical, observational study examined 11,722 hospital admissions of adult patients in 2019, each with a minimum hospital stay of 48 hours (h) in a tertiary hospital. Rates and timing for the detection, referral, and assessment of patients at nutritional risk were calculated. Participants were divided into low (Malnutrition Screening Tool [MST] < 2 points) and moderate/high (MST ≥ 2) nutritional risk groups to evaluate the relationship between nutritional risk and clinical and demographic variables.

Results: We found that 91% of patients underwent nutritional screening within the first hours of admission, with a median time of 9 h from admission to screening (interquartile range [IQR] 3–19). The prevalence of nutritional risk (MST ≥ 2) was 21%. All patients identified as being at nutritional risk were immediately referred for a nutritional assessment once identified, with a median referral time of 0 h (IQR 0–0). This assessment was carried out by a nutritionist for 98% of these patients, with a median time of 19 h from referral to assessment (IQR 6–24). Compared to the low-risk group, patients with nutritional risk were older, had higher rates of mortality and admission to the intensive care unit (ICU), longer hospital stays, and a higher proportion of men and cancer diagnoses ($p < 0.001$ for all comparisons). After adjusting for age and sex, nutritional risk was significantly associated with a higher probability of ICU admission (Odds Ratio [OR] 1.13; 95% CI 1.02–1.24) and in-hospital mortality (OR 2.32; 95% CI 1.97–2.73).

Conclusion: The integration of technology into nutritional screening was highly efficient for early detection and assessment of at-risk patients upon hospital admission. Features of this system could guide other hospitals. The association found between nutritional risk and clinical outcomes emphasizes the importance of prompt and appropriate nutritional interventions.

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* Corresponding author. Departamento de Alimentación y Nutrición, Facultad de Ciencias de la Salud, Pontificia Universidad Javeriana Seccional Cali, Cali 760021, Colombia.

** Corresponding author.

E-mail addresses: frank.carrera@javerianacali.edu.co (F. Carrera-Gil), maria.prieto@fvli.org.co (M.I. Prieto Rusca).

¹ Present address: Departamento de Alimentación y Nutrición, Facultad de Ciencias de la Salud, Pontificia Universidad Javeriana Seccional Cali, Cali 760021, Colombia.

1. Introduction

Malnutrition, commonly known as disease-related malnutrition, represents a critical issue in hospitals, affecting approximately 20–50% of hospitalized patients worldwide. This nutritional imbalance leads to various adverse health outcomes, including immune system dysfunction, muscle atrophy, delayed wound healing, and increased healthcare costs and mortality rates [1].

Nutritional screening is a simple procedure to identify individuals at nutritional risk and represents a crucial initial step in managing this issue. 'Nutritional risk' is defined as the likelihood of experiencing negative clinical outcomes due to nutritional factors, which could be modified through appropriate nutritional therapy [2]. In this context, patients with moderate or high nutritional risk are more likely to experience serious health issues compared to those with low nutritional risk, highlighting the importance of implementing nutritional therapy strategies.

Indeed, experimental research has shown that early identification and management of patients at nutritional risk can significantly reduce mortality, complications, and hospital readmissions [3–6]. Therefore, in accordance with the Joint Commission International Standards for patient safety [7], it is imperative to implement early detection and treatment strategies to minimize the negative impacts of malnutrition in the hospital setting.

However, despite the evidence, the systematic implementation of nutritional screening in hospitals remains a considerable challenge. International reports reveal low rates of detection and/or early intervention in at-risk patients [8], a situation that persists even in countries where nutritional screening is mandatory [9]. The main obstacles include time constraints, workloads, lack of education and training, and absence of guidelines or local policies [10].

To overcome these challenges, the Clinical Nutrition Service of a tertiary hospital in Colombia has implemented a technology-assisted nutritional screening system. The aim of this study is to evaluate the efficiency of this system in early detection and assessment of patients upon hospital admission. Additionally, it explores how nutritional risk identified at hospital admission is associated with clinical outcomes and sociodemographic characteristics of patients.

2. Materials and methods

2.1. Study design

In a retrospective analytical observational study conducted at a tertiary hospital, all adult patient admissions from the year 2019 were retrieved. Each hospital admission was counted as a new patient, even in cases of multiple hospitalizations of the same individual. Individuals aged 18 years and older were considered for inclusion in the study, but those with hospital stays of less than 48 h and those without a record of nutritional screening were excluded. The study received approval from the Biomedical Research Ethics Committee of the hospital, as documented in approval certificate Number 005-202.

2.2. Description and assessment of the nutritional screening system

In 2015, the Clinical Nutrition service led the development of a technology application for nutritional risk management, integrating the Malnutrition Screening Tool (MST) into our institution's electronic medical record (EMR). We chose the MST because it is a simple, fast, reliable, and valid tool that can be used with minimal training by any healthcare staff member. It inquires about recent involuntary weight loss and decreased dietary intake due to lack of appetite, stratifying nutritional risk into two categories: low risk (score less than 2) and moderate or high risk (score equal to or greater than 2).

The application that integrates the MST into the EMR was developed by our hospital's computer engineers using SAP®, the informatics platform that supports our EMR system. This application imports general patient information from the EMR, such as full name, age, sex, personal identification number, bed number, and a numeric identifier for the hospitalization. It allows the examiner to

select responses for each question of the MST without the need to input free text, thus minimizing typographical errors. Additionally, it automatically categorizes patients into low or moderate/high nutritional risk based on the MST scoring system.

SAP software was chosen for its robust, reliable, and integrative capabilities, which are crucial for managing comprehensive medical records. It is also known for its strong data security features, user-friendly interface, and ability to seamlessly integrate various healthcare applications. This ensures that the nutritional screening process is efficient, accurate, and easily accessible for all healthcare staff.

To guarantee the systematic application of the MST upon hospital admission and weekly during hospitalization, the nutritional risk system was implemented with five key functionalities:

- 1. Integration of MST into a specific routine nursing procedure:** Consequently, it was linked to the Braden scale, a tool that nursing staff use at every work shift to assess the risk of developing pressure ulcers in all hospitalized patients.
- 2. Emission of reminders:** Both at the time of patient admission and every seven days after the initial screening. When nursing staff record the Braden Scale for the first-time during hospitalization, the system verifies whether a nutritional screening has been previously conducted. If the nutritional screening has already been completed, the registration of the Braden scale is allowed. However, if the screening has not yet been done, the system displays a visual alert indicating the need to complete the screening before recording the Braden Scale.
- 3. Automatic referral of patients at risk:** Patients with moderate or high risk are automatically referred to the Clinical Nutrition service, without the need for phone calls or medical orders. The nutritional risk system includes a dashboard where patients at-risk are displayed once detected, allowing nutritionists to track them in real-time.
- 4. Full registration of screening data:** This includes responses to each question, risk score, time and date of registration, medical service where it was applied, bed number, full name, and age of the patient.
- 5. Tracking of nutritional assessment:** The system tracks the nutritional assessment of patients identified as at nutritional risk.

In 2023, following years of ongoing evaluation that identified areas for improvement, three key functions were added to the nutritional risk system. These enhancements were designed to reduce nursing workload, enhance data integrity, and strengthen the diagnostic accuracy of screening:

1. Identification of patients receiving nutritional care during hospitalization and elimination of the need for screening in these patients.
2. Inability to modify screening records, but allowing for new registration in case of errors in the application process.
3. Statistics on false positives identified by the nutritionist.

To evaluate the screening system's efficiency in detecting and early assessment of patients at nutritional risk, six essential metrics were calculated to comprehensively understand its performance:

1. Patients with a nutritional screening upon hospital admission.
2. Patients at risk referred for a nutritional assessment.
3. Patients at risk assessed by a nutritionist.
4. Time from admission to the first nutritional screening.
5. Time from identification to referral of at-risk patients.
6. Time from referral to assessment of at-risk patients

2.3. Data collection techniques

Nutritional screening data for all participants were extracted from the nutritional risk system. Clinical and sociodemographic variables were obtained from electronic medical records of patients who met the selection criteria. Both nutritional screening variables and clinical/sociodemographic data were recorded in a secure database managed by the researchers.

2.4. Statistical analysis

The distribution of continuous variables was assessed using the Kolmogorov-Smirnov test, given that the number of participants exceeded fifty [11]. Mean and standard deviation are presented for variables that followed a normal distribution. For variables not normally distributed, median and interquartile range (IQR) are reported. For categorical variables, absolute frequencies (number of cases) and relative frequencies (proportions relative to the total cases) were calculated.

To assess the relationship between nutritional risk and socio-demographic and clinical variables, participants were categorized into two groups: those with low nutritional risk ($MST < 2$) and those with medium or high nutritional risk ($MST \geq 2$). Subsequently, Mann-Whitney tests were used to compare these groups for continuous variables that did not follow a normal distribution, while Chi-square test or Fisher's exact tests was employed for categorical variables.

Additionally, a binary logistic regression analysis was conducted, with the level of nutritional risk as the independent variable and clinical outcomes (admission to intensive care unit and in-hospital mortality) as dependent variables. The regression model was adjusted for age and sex as covariates, and the resulting associations were presented in terms of odds ratios (OR) along with 95% confidence intervals. This methodological approach allowed for a rigorous examination of the relationship between nutritional risk and clinical outcomes, adequately controlling for sociodemographic factors, and provided quantitative results to understand the magnitude of these associations. Statistical significance was set at a probability to reject the null hypothesis of < 0.05 .

3. Results

In 2019, a total of 19,566 hospital admissions of adult patients were identified. 7844 did not meet the selection criteria, resulting in a total of 11,722 admissions eligible for analysis (Fig. 1). The clinical and sociodemographic characteristics of the analyzed population are shown in Table 1. Participants had a median age of 56 years (IQR 36–69), with 55% being women. The most common medical category was hemato-oncology, occurring in 31% of cases.

Compared to individuals classified as low-risk, those at nutritional risk exhibited several significant differences. They were notably older (60 vs 54 years, $p < 0.001$), experienced a higher rate of in-hospital mortality (11% vs 5%, $p < 0.001$), were more frequently admitted to the intensive care unit (35% vs 31%, $p < 0.001$), and had a greater proportion of men (51% vs 49%, $p < 0.001$). Moreover, the nutritional risk group showed a higher prevalence of malignant diseases (40% vs 29%, $p < 0.001$) and had a longer median hospital stay, with 7 days compared to 6 days for the low-risk group ($p < 0.001$). Additionally, there was a higher proportion of patients with contributory and subsidized health insurance regimes in the nutritional risk group ($p < 0.001$).

Table 2 illustrates the efficiency of identifying and assessing the nutritional status of patients at risk. Specifically, 91% of patients (11,722 out of 12,917) underwent nutritional screening within the

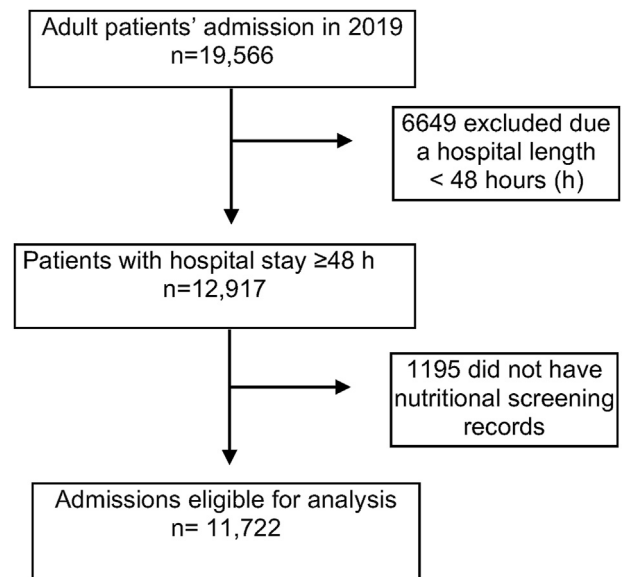


Fig. 1. Participant flow.

first few hours of admission, with a median time of 9 h from admission to screening (IQR 3–19).

Upon hospital admission, nutritional risk, defined as an MST score of ≥ 2 , was identified in 21% of participants. Following identification, nutritionists of the Clinical Nutrition Service conducted a documented assessment for 98% of these at-risk patients, within a median time of 19 h (IQR 6–24). Nutritional assessment was performed using standardized procedures that led to a nutritional diagnosis and planning of individualized interventions.

Table 3 shows the relationship between nutritional risk at admission, hospital mortality, and admission to the ICU. Patients at nutritional risk were 1.22 times more likely to be admitted in the ICU compared to low-risk patients (95% CI 1.12–1.35). This association remained significant after adjusting for age and sex (OR 1.13; 95% CI 1.02–1.24). Regarding hospital mortality, the probability of death was 2.58 times higher in patients with nutritional risk (95% CI 2.2–3.03), decreasing to 2.32 after adjustment, but still maintaining statistical significance (95% CI 1.97–2.73).

4. Discussion

We found high efficiency in our screening system for the early detection and assessment of patients at nutritional risk. These results highlight the importance and benefits of integrating technology into nutritional screening procedures.

This evidence is reinforced when comparing our findings with studies conducted in similar medical contexts and populations. The study by Cardenas et al., based on data from seven years of the Nutrition Day survey in Colombia, revealed that only 50% of patients underwent nutritional screening upon hospital admission, and only 23% of at-risk patients received nutritional intervention [12]. Although Colombia has a policy of patient safety that includes tackling hospital malnutrition [13], the effective management of nutritional risk continues to be a significant challenge in many of its institutions. This challenge is not exclusive to Colombia but is also observed in hospitals in developed countries, indicating its universal nature.

In a study conducted in 90 hospitals in the United States, Sherry et al. reported that 90% of patients underwent a screening process within the first 24 h of admission. However, it is important to note

Table 1
Participant characteristics and group comparison by nutritional risk level.

Characteristics	All patients n = 11,722	Low risk n = 9305	Medium/high risk n = 2417	P value
Age, median (IQR)	56 (36–69)	54 (35–68)	60 (43–72)	<0.001
Sex, n (%)				
Female	6392 (55)	5210 (56)	1182 (49)	<0.001
Male	5330 (45)	4095 (44)	1235 (51)	
Health insurance, n (%)				
Contributive	5675 (48)	4351 (47)	1324 (55)	<0.001
Subsidized	1889 (16)	1479 (16)	410 (17)	
Prepaid	1349 (12)	1143 (12)	206 (9)	
Particular	2809 (24)	2332 (25)	477 (20)	
Unit service, n (%)				
General ward	8009 (68)	6444 (69)	1565 (65)	<0.001
ICU	3713 (32)	2861 (31)	852 (35)	
Medical category, n (%)				
Hemato-Oncology	3699 (31)	2730 (29)	969 (40)	<0.001
Genitourinary	3536 (30)	1022 (11)	260 (11)	
Cardio-metabolic	1479 (13)	1181 (13)	298 (12)	
Digestive	1010 (9)	787 (8)	223 (9)	
Respiratory	288 (2)	498 (5)	156 (6)	
Other	1710 (15)	3087 (33)	511 (21)	
Hospital length (days), median (IQR)	6 (4–12)	6 (3–11)	7 (4–15)	<0.001
In-hospital mortality, n (%)	693 (6)	426 (5)	267 (11)	<0.001

ICU, intensive care unit; IQR, interquartile range.

Table 2
Efficiency in the detection and evaluation of patients at nutritional risk.

Nutritional Screening	
Screened patients at admission, n (%)	11,722 (91)
Time from admission to screening, median hours (IQR)	9 (3–19)
Patients at nutritional risk (MST $\geq$ 2), n (%)	2417 (21)
Nutritional Assessment	
Patients at risk referred for a nutritional assessment, n (%)	2417 (100)
Time to refer patients at risk, median hours (IQR)	0 (0–0)
At-risk patients assessed by the nutritionist, n (%)	2362 (98)
Time to nutritional assessment of patients at risk, median hours (IQR)	19 (6–24)

IQR, interquartile range; MST, malnutrition screening tool.

that only half of these institutions utilized a validated screening tool, and only 60% of at-risk patients received appropriate interventions [14]. Similar data were reported by Eglseer et al. in a study that covered 53 hospitals in Austria. In this case, it was found that 74.5% of patients were assessed upon admission, but only 21.2% of these cases used a validated screening tool [15].

In Portugal, screening for nutritional risk is mandatory in public hospitals; however, Taipa-Mendes et al. reported that the implementation rate of nutritional screening was only 25.4% $\pm$ 2.2 in 41 hospitals, despite having a screening tool integrated into the electronic medical record [9]. The authors identified various barriers to conducting screenings, such as staffing shortages and lack of knowledge about this practice. This emphasizes the importance of a well-integrated and automated nutritional risk system, like the one we present, which overcomes these limitations.

Table 3
Association of medium/high nutritional risk at admission with clinical outcomes.

Variable	OR	95% CI	OR adjusted	95% CI
Admission to ICU	1.22	1.12–1.35	1.13	1.02–1.24
In-hospital mortality	2.58	2.2–3.03	2.32	1.97–2.73

Participants classified as low risk at admission served as the reference group for establishing the association between medium/high nutritional risk and clinical outcomes. OR, odds ratio; ICU, intensive care unit. CI, confidence interval. OR adjusted by age and sex.

In Singapore, Lim et al. conducted audits at a tertiary-level hospital and found an error rate of 33% in nutritional risk detection, as well as significant delays in referral to the dietitian, with an average delay of 4.3 $\pm$ 1.8 days. These findings led to the implementation of a continuous improvement project that yielded excellent results, including referrals made by nursing staff instead of physicians [16]. In our hospital, we have successfully automated the referral process entirely, achieving a universal and immediate referral rate for all identified cases at nutritional risk, alleviating the workload of other healthcare professionals.

These international comparisons point out the significant advancements we have achieved with our approach to nutritional risk detection and assessment. The combination of technology, automated processes, and continuous training has greatly enhanced the efficiency of nutritional screening, ensuring faster assessments and interventions, which makes a significant difference in the quality of hospital care. This is consistent with a narrative review [17] showing that computerized applications integrated into hospital information systems have improved the early detection of patients at nutritional risk [18–21].

Nutritional screening offers well-established clinical and economic benefits [22]; nevertheless, it is crucial to understand how this process can lead to excessive workload due to false positive cases. For the MST tool, the false positive rate in hospital settings ranges from 5% to 29% [23,24]. Additionally, it is reported that nutritionists spend 15–20 min investigating each false positive [23]. Moreover, this consumes valuable time that could be spent attending to actual at-risk cases. We addressed this issue by eliminating the need to document the assessment of these cases once identified, allowing nutritionists to record them as false positives directly in the nutritional risk application. Furthermore, continuous training for all nursing staff has been intensified, focusing on areas with a high incidence of false positives. These measures help minimize redundant work and enhance the overall quality of the screening process.

The patients identified as being at nutritional risk in our study had a longer hospital stay and a higher likelihood of admission to ICU, as well as an increased risk of mortality. These findings are consistent with other observational studies [25–28] and with the body of experimental research [29,30], underscoring the crucial

role of nutritional screening as a simple and cost-effective tool for risk stratification and clinical management. Additionally, among our patients identified as being at nutritional risk, a higher proportion of men and individuals with cancer diagnoses were noted. They also exhibited a significantly older age, with a median age of 60 years. In a large observational study, both advanced age and cancer have been associated with malnutrition [31]. This correlation could be attributed to increased comorbidity, functional decline, and metabolic abnormalities prevalent in these populations. Regarding sex, it's noteworthy that certain chronic diseases, such as cancer, disproportionately affect men compared to women [32]. This discrepancy may contribute to a higher incidence of malnutrition among male patients in hospital settings. Understanding these demographic and disease-specific factors is crucial for effective nutritional assessment and for developing intervention strategies tailored to each patient's needs.

Our approach to nutritional care recognizes that patients' body composition and functionality depend on more than just early intervention and adequate nutrient intake. Reduced mobility, common in hospitalized patients, can lead to significant muscle mass loss in a few days, even with controlled nutrition [33]. Additionally, systemic inflammation can affect body composition [34] and undermine the benefits of nutritional therapy in patients at nutritional risk [35]. These considerations stress the need for a comprehensive nutritional approach that goes beyond mere risk identification and assessment, prioritizing continuous nutrition and proper management of other critical factors.

5. Limitations and strengths

An important limitation of our study is the lack of a control group, which restricts the direct assessment of the clinical effects of our strategies. Furthermore, our reliance on retrospective data means that we cannot account for all potential confounders, such as underlying comorbidities or variations in clinical practice that could affect outcomes. While the MST tool is validated, its accuracy might vary depending on staff training levels and adherence to protocol. Additionally, the single-center design of our study may limit the generalizability of our findings to other settings. Future research should aim to address these limitations by including control groups, collecting prospective data, and engaging in multi-center studies to enhance the robustness and applicability of the findings.

A significant strength of our research is the large number of participants. This has allowed us to obtain estimates with narrower confidence intervals, reinforcing the robustness of our conclusions. Additionally, we believe that our work is among the first to detail the essential steps for successfully developing and implementing a nutritional screening system. We hope that the information provided in our study will not only serve as a guide for other institutions seeking to implement similar practices but also be of interest to scientific societies dedicated to promoting high-quality nutritional care.

6. Conclusion

A nutritional screening system that integrates technology was highly efficient in identifying and evaluating patients at nutritional risk within a complex clinical setting. The observed association between nutritional risk and clinical outcomes highlight the critical importance of timely and suitable nutritional interventions. To enhance our understanding, prospective studies are warranted to examining the clinical impacts resulting from the implementation of this screening system.

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Authors contributions

FCG: Conceptualization, Methodology, Writing - Original Draft, Data Curation, Formal analysis Writing - Review & Editing. **MIPR:** Writing - Original Draft, Formal analysis, Investigation, Visualization. All authors critically reviewed and approved the final manuscript.

Declaration of competing interest

None.

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